Calculus I

Computation problems #8

Topics:

- Volume using cross-sections
- Volume using cylindrical shells
- Arc length
- Areas of surfaces

Problem 1. Find the volume of the solid generated by revolving the region bounded by the curve $y = 4 - x^2$ and the line $y = 2 - x$ about the x-axis.

Problem 2. Find the volume of the solid generated by revolving the region in the first quadrant bounded above by the curve $y = x^2$, below by the x-axis and on the right by the line $x = 1$, about the line $x = -1$.

In Problems 3–4, do the following.

(a) Set up an integral for the length of the curve.

(b) Graph the curve to see what it looks like.

(c) Use your graphers or computers integral evaluator to finf the curves length numerically.

Problem 3. $y^2 + 2y = 2x + 1$ from (-1, -1) to (7, 3).

Problem 4. $y = \displaystyle\int_0^x \tan t \, dt, \quad 0 \le x \le \pi/6.$

In Problems 5–6, do the following.

(a) Set up an integral for the area of the surface generated by revolving the given curve about the indicated axis.

(b) Graph the curve to see what it looks like. If you can, graph the surface too.

(c) Use your utility's integral evaluator to find the surface's area numerically.

Problem 5. $x^{1/2} + y^{1/2} = 3$ from (4, 1) to (1, 4); x-axis.

Problem 6. $y = \displaystyle\int_1^x \sqrt{t^2 - 1}\, dt, \quad 1 \le x \le \sqrt{5}$; x-axis.

Find the areas of the surfaces generated by revolving the curves in Problems 7–8 about the indicated axes.

Problem 7. $y = \sqrt{2x - x^2}, \quad 10.5 \le x \le 1.5$; x-axis.

Problem 8. $x = \sqrt{2y - 1}, \quad 15/8 \le y \le 1$; y-axis.

Problem 9.

(a) Show that for $0 < \alpha \le \pi/2$,

$$\int_0^\alpha \sqrt{1 + \cos^2\theta} d\theta > \sqrt{\alpha^2 + \sin^2\alpha}.$$

(b) Generalize the result in part (a).

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Answer to selected problems

1. $\dfrac{108\pi}{5}$
3. ≈ 9.29
4. (a) and (c) ≈ 0.549
5. ≈ 63.366
8. ≈ 2.997